

Lecture Scribe: Inequalities and Tail Bounds

1 Tail of a Random Variable and Tail Probability

1.1 Definition

The tail of a random variable X is defined as:

$$P(X > x)$$

This represents the probability that the random variable takes a value greater than a threshold x .

2 Motivation for Studying Tail Events

2.1 Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

These represent rare but critical events.

2.2 Key Idea

- Mean $\rightarrow$ average behavior (easy to analyze)
- Tail $\rightarrow$ rare extreme events (hard to analyze)

2.3 Why are tails hard to compute?

- Requires summing many small probabilities

3 Exact Computation of Tail Probabilities

3.1 Example 1: Binomial Case

Let:

$$X \sim \text{Binomial}(n, p)$$

We want:

$$P(X \geq k)$$

Exact computation:

$$P(X \geq k) = \sum_{i=k}^n \binom{n}{i} p^i (1-p)^{n-i}$$

3.2 Example 2: Poisson Case

Let:

$$X \sim \text{Poisson}(\lambda)$$

Then:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

3.3 Observation

- These are long, complicated summations
- Hence exact computation is difficult

4 Idea of Tail Bounds

4.1 Motivation

Instead of computing exact values:

- Find an upper bound
- Easier to compute
- Still provides a guarantee

$$P(X \geq k) \leq (\text{simple expression})$$

4.2 Key Idea

Approximate safely instead of computing exactly.

5 Running Example

- Flip a fair coin n times
- Let X = number of heads

Distribution:

$$X \sim \text{Binomial}(n, 1/2)$$

Mean:

$$E[X] = \frac{n}{2}$$

Tail event:

$$P\left(X \geq \frac{3n}{4}\right)$$

Interpretation:

This corresponds to getting unusually many heads.

6 Markov's Inequality

6.1 Assumptions

$$X \geq 0$$

6.2 Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

6.3 Key Idea

- Uses only the mean
- Large values are rare

6.4 Proof

1. Write expectation:

$$E[X] = \sum_x x \cdot P(X = x)$$

2. Restrict to values where $X \geq a$:

$$E[X] \geq \sum_{x \geq a} x \cdot P(X = x)$$

3. Since $x \geq a$:

$$E[X] \geq \sum_{x \geq a} a \cdot P(X = x)$$

4. Factor out a :

$$E[X] \geq a \cdot P(X \geq a)$$

5. Rearranging:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

6.5 Example

If:

$$E[X] = 10$$

Then:

$$P(X \geq 100) \leq \frac{10}{100} = 0.1$$

7 Chebyshev's Inequality

7.1 Assumptions

Mean μ , variance σ^2

7.2 Statement

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

7.3 Derivation

1. Let:

$$Y = (X - \mu)^2 \geq 0$$

2. Apply Markov:

$$P(Y \geq k^2) \leq \frac{E[Y]}{k^2}$$

3. Since:

$$E[Y] = \sigma^2$$

4. Hence:

$$P((X - \mu)^2 \geq k^2) \leq \frac{\sigma^2}{k^2}$$

5. Equivalent to:

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

8 Chernoff Bound

8.1 Goal

Provide exponentially tighter bounds.

8.2 Key Idea

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Apply Markov:

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Thus:

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

8.3 Optimization

Choose t to minimize the bound.

9 Poisson Tail Bounds

For:

$$X \sim \text{Poisson}(\lambda)$$

Exact:

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Instead, use Chernoff bounds.

10 Jensen's Inequality

10.1 Definition

A function f is convex if:

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

10.2 Statement

$$f(E[X]) \leq E[f(X)]$$

10.3 Example

If $f(x) = x^2$:

$$(E[X])^2 \leq E[X^2]$$

11 Connections

11.1 Core Flow

1. Exact tails are hard
2. Use inequalities:
 - Markov (mean)
 - Chebyshev (variance)
 - Chernoff (exponential)
3. Goal:

$$P(X \geq k) \leq \text{simple bound}$$

11.2 Hierarchy

- Markov $\rightarrow$ weakest
- Chebyshev $\rightarrow$ stronger
- Chernoff $\rightarrow$ strongest

12 Final Takeaways

- Tail probability: $P(X > x)$
- Exact computation is difficult
- Inequalities provide upper bounds

- Markov $\rightarrow$ mean
- Chebyshev $\rightarrow$ variance
- Chernoff $\rightarrow$ exponential bounds
- Jensen connects convexity and expectation

Bound rare events instead of computing them exactly